

MedImageInsight: Comprehensive Benchmark & Accuracy Findings

Reference Study: *MedImageInsight: An Open-Source Embedding Model for General Domain Medical Imaging*

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Document Scope: Complete extraction and synthesis of all empirical results, accuracy benchmarks, model comparisons, and clinical findings.

Executive Summary & Core Discoveries:

- **SOTA / Expert Equivalence with Sub-Billion Footprint:** With a 360M image encoder (DaViT) and 252M text encoder (<0.61B total parameters), MedImageInsight achieves SOTA or SOTA-equivalent performance in 5 public benchmark tasks, surpasses human expert accuracy in 2 tasks (Bone Age error: 6.19 vs 7.32 months), and maintains mAUC > 0.90 across multi-domain medical modalities.
- **Zero-Penalty Explainability (KNN Search):** Evaluated using k-Nearest Neighbor (KNN) image-image retrieval, predictions deliver transparent evidence-based reasoning with minimal to no accuracy loss compared to black-box softmax classifiers.
- **Regulatory Compliance (True ROC Curves):** The model outputs continuous probability distributions, providing true ROC curves and sensitivity/specificity calibration necessary for FDA SaMD certification.
- **Compute & Parameter Efficiency:** For radiology report generation, pairing the model with a 70M parameter text decoder (0.4B total) achieves clinical efficacy (CheXbert F1: 56.3–57.9) matching or exceeding 7B–84B parameter systems (MAIRA-1, LLaVA-Rad, Med-PaLM M).
- **Demographic Equity:** Validated on 1,780 independent hospital patients with consistent performance across gender (0.3% gap) and age cohorts (≤20 to ≤100 years).

1. Medical Foundation Model Capabilities Benchmark

Comparison of architectural and clinical capabilities between MedImageInsight and published medical foundation models.

Model	Params	Multi-Domain	Text Pretrain	Label Pretrain	ROC (Regulatory)	Image Search	Generative	3D Retrieval
CXR Foundation	0.25B	✗	✗	✓	✓	✗	✗	✗
ELIXR	0.25B	✗	✓	✓	✓	✗	✓	✗
BiomedCLIP	0.09B	✓	✓	✗	✓	✗	✓	✗
MAIRA-1 / 2	7–13B	✗	✓	✗	✗	✗	✓	✗
LLaVA-Med / Rad	7B	✓ / ✗	✓	✗	✗	✗	✓	✗
Med-Gemini	>3.25B	✓	✓	✓	✗	✗	✓	✗
Med-PaLM-M	84–562B	✓	✓	✓	✗	✗	✓	✗
BiomedGPT	0.18B	✓	✓	✗	✓	✗	✓	✗
MedImageInsight	<0.61B	✓	✓	✓	✓	✓	✓	✓

2. Comprehensive Multi-Domain Benchmark Results

Empirical results across 14 medical imaging modalities. Compares zero-shot text-prompt classification against KNN image-image search and state-of-the-art baselines.

Modality	Sub-Modality	Dataset (# Labels)	Metric	MI2 (Zero-Shot)	MI2 (KNN Search)	SOTA / Ref.	Finding / Outcome
Radiology	X-Ray (CXR)	LT-CXR (20)	mAUC / BACC	0.853 / 0.422	0.791 / 0.334	— / 0.294	+12.8% BACC vs SOTA
Radiology	X-Ray (CXR)	MGB CXR (80)	mAUC	0.937	0.839	—	High discriminative power
Radiology	X-Ray	IRMA 2009 (137)	mAUC	0.989	0.915	—	Excellent anatomy/view coding
Radiology	X-Ray	RSNA Bone Age (-)	Abs. L1 (mo)	—	6.194 mo	7.32 mo	Exceeds human expert (7.32 mo)
Radiology	X-Ray	MGB Bone Age (-)	Abs. L1 (mo)	—	6.570 mo	7.32 mo	Exceeds human expert on clinical test
Radiology	X-Ray	RSNA Mammo. (4)	mAUC	0.934	0.843	—	Breast cancer finding triage
Radiology	CT	LIDC-IDRI (4)	mAUC	0.767	0.718	—	Nodule malignancy grading
Radiology	MR / CT	Exam Parameter (21)	mAUC	0.977	0.958	—	View/protocol identification
Radiology	Ultrasound	BUSI (3)	mAUC	0.985	0.977	—	Breast lesion differentiation
Radiology	Ultrasound	HMC-QU View (2)	mAUC	0.987	0.970	—	2CH vs 4CH echocardiography
Radiology	Ultrasound	Bing Echo View (7)	mAUC	0.941	0.946	—	KNN outperforms text classification
Dermatology	Dermoscopy	ISIC 2019 (8)	mAUC	0.966	0.852	0.954	Exceeds published SOTA (0.954)
Dermatology	Clinical Photo	SD-198 (198)	mAUC / ACC	0.989 / 0.696	0.952 / 0.700	— / 0.578	+12.2% Accuracy vs SOTA (70% vs 58%)
Dermatology	Clinical Photo	PAD-UFES-20 (6)	mAUC	0.954	0.929	0.973*	*Med-PaLM M used 14 clinical traits
Pathology	Histology H&E;	PCAM (2)	mAUC	0.963	0.943	0.975	Lymph node metastasis detection
Ophthalmology	OCT	OCT2017 (4)	mAUC / ACC	1.000 / 1.000	1.000 / 1.000	0.998 / 0.952	Flawless 100% mAUC & Accuracy
Ophthalmology	OCT	OCT2018 (4)	mAUC / ACC	0.999 / 0.975	0.990 / 0.936	— / 0.909	+6.6% Accuracy vs SOTA (97.5% vs 91%)
Ophthalmology	Retinal Fundus	ODIR-5K (28)	mAUC	0.949	0.803	—	Multi-disease fundus triage

3. Fine-Tuning & Report Generation Benchmarks

Table 4: Downstream Task Fine-Tuning on ChestX-ray14

Evaluates whether a multi-domain foundation model matches models trained exclusively on chest radiographs.

Model	Pretrain Domains	Model Size	Edema (AUC)	Consolidation (AUC)	Effusion (AUC)
CXR Foundation	1 (CXR only)	240M	0.827	0.727	0.813
ELIXR	1 (CXR only)	240M	0.856	0.749	0.835
MedImageInsight	14 (General Domain)	360M	0.860	0.744	0.832

Table 5: Radiology Findings Report Generation on MIMIC-CXR

Evaluated on official MIMIC-CXR test split using clinical efficacy (CheXbert, RadGraph) and lexical metrics (BLEU, ROUGE).

Model Category	Model Name	Total Size	CheXbert Micro F1-14	CheXbert Micro F1-5	CheXbert Macro F1-14	RadGraph F1	BLEU-1	BLEU-4	ROUGE-L
Models > 7B	LLaVA-Rad (Frontal)	7B	57.3	57.4	39.5	—	38.1	15.4	30.6
Models > 7B	Med-PaLM M	84B	53.6	57.9	39.8	26.7	32.3	11.3	27.3
Models > 7B	MAIRA-1 (Frontal)	7B	55.7	56.0	38.6	24.3	39.2	14.2	28.9
Models > 7B	CheXagent	8B	39.3	41.2	24.7	—	16.9	4.7	21.5
Models > 7B	GPT-4o (finetuned)	>100B	48.9	52.7	33.0	26.4	39.6	17.8	32.1
Models < 1B	CvT2Dist	0.3B	44.2	—	30.7	—	39.3	12.7	28.6
Models < 1B	R2Gen	0.2B	22.8	34.6	—	—	35.3	8.6	27.7
Ours (<0.5B)	MedImageInsight	0.4B	53.6	57.2	37.0	25.5	37.0	15.2	31.6
Ours (<0.5B)	MedImageInsight (Frontal)	0.4B	56.3	57.9	38.4	25.4	37.3	15.3	31.7
Ablation	Ablative Baseline (no pretrain)	0.4B	49.1	53.9	31.0	23.6	33.5	13.4	29.9

4. Independent Real-World Clinical Evaluation

Evaluated on real-world, unseen patient data at a partner clinical institution using KNN image-image classification across acute ICU findings and tubes.

Model	ETT (AUC)	NGT (AUC)	Pneumo-peritoneum	Pneumo-thorax	Rib Fracture	Vascular Lines	Normal (AUC)	Multi-Class mAUC	Binary Abnormal
Med-Flamingo	67.2	71.1	66.3	68.8	86.3	67.0	83.3	72.9	83.4
BiomedCLIP	79.2	80.0	73.6	75.5	89.8	71.5	90.2	80.0	88.5
RAD-DINO	82.6	82.7	80.5	78.1	91.5	81.1	90.5	83.9	90.5
MedImageInsight	87.5	89.7	89.6	91.3	95.1	88.8	96.7	91.2	96.7

5. 3D Medical Image Retrieval (3D-MIR Benchmark)

Evaluates 3D CT volume search across four abdominal and thoracic organs (Liver, Pancreas, Lung, Colon) for Tumor Presence and Tumor Staging.

Organ / Target	Model	Tumor P resence P@3	Tumor P resence P@5	Tumor P resence P@10	Tumor P resence AvgPrec	Tumor Stage P@3	Tumor Stage P@5	Tumor Stage P@10	Tumor Stage AvgPrec
Liver	Med-Flamingo	42.1	45.3	47.7	50.6	35.1	35.8	36.8	41.4
Liver	BiomedCLIP	79.0	77.9	76.9	81.6	40.4	43.2	44.6	50.2
Liver	BiomedGPT	82.5	81.0	79.6	84.0	45.6	49.5	46.1	55.4
Liver	MedImageInsight	80.7	77.9	81.7	82.3	43.9	42.1	44.5	54.6
Pancreas	BiomedGPT	100.0	100.0	99.4	100.0	44.8	51.9	57.8	61.1
Pancreas	MedImageInsight	100.0	100.0	99.4	99.9	60.4	58.1	57.6	67.3
Lung	BiomedCLIP	99.3	98.3	98.3	99.1	73.7	65.1	61.8	83.1
Lung	MedImageInsight	100.0	100.0	96.9	99.7	69.8	73.8	68.6	76.1
Colon	BiomedCLIP	84.7	83.3	78.8	87.1	51.4	51.7	47.9	63.3
Colon	MedImageInsight	100.0	99.2	97.3	99.7	65.3	62.5	60.6	69.4
Macro Average	Med-Flamingo	72.0	73.8	72.3	78.1	51.7	51.4	48.3	59.3
Macro Average	BiomedCLIP	90.2	89.3	87.7	91.4	53.4	52.6	52.7	64.6
Macro Average	BiomedGPT	85.5	83.4	80.4	91.1	53.1	55.2	53.6	63.1
Macro Average	MedImageInsight	95.2	94.3	93.8	95.4	59.8	59.1	57.8	66.9

6. AI Demographic Fairness & Bias Assessment

Detailed demographic stratification across 1,780 patients from the independent clinical partner hospital.

Demographic Cohort	Sample Size (n)	MedImageInsight mAUC	MI2 Binary Abnormal	RAD-DINO mAUC	BiomedCLIP mAUC	Med-Flamin go mAUC
Female Patients	709 patients	92.5%	96.8%	84.8%	82.3%	74.6%
Male Patients	1,059 patients	90.4%	96.5%	83.1%	78.5%	71.8%
Age: ≤ 20 years	164 patients	94.1%	99.7%	86.1%	86.3%	85.5%
Age: ≤ 40 years	237 patients	89.8%	97.6%	83.8%	83.6%	74.8%
Age: ≤ 60 years	570 patients	91.0%	97.7%	84.9%	79.4%	73.5%
Age: ≤ 80 years	610 patients	90.9%	93.6%	83.2%	78.0%	69.5%
Age: ≤ 100 years	99 patients	92.1%	98.6%	79.2%	76.7%	72.0%

7. Selected Pathology Highlights & Technical Architecture

Model Architecture & Training Details:

- **Image Encoder:** Dual Attention Vision Transformer (DaViT), 360 Million parameters.
- **Language Encoder:** Transformer text encoder, 252 Million parameters.
- **Total Embedding Model Size:** <0.61B parameters. At inference, text prompts are encoded once to create an instantaneous classification head.
- **Training Objective:** Unified Contrastive Learning (UniCL) over mixed text-label supervision.
- **Pre-training Compute:** 7,680 V100 GPU hours (64 NVIDIA V100 GPUs on Azure Machine Learning, West US 2), total carbon footprint of 0.8918 tCO2eq.
- **Total Pre-training Scale:** 3,786,210 images across 26 distinct medical datasets (MIMIC-CXR, PMC-15M, LIDC-IDRI, PatchCamelyon, MSD, SD-198, ISIC, etc.).

Category-Level AUC Highlights from Appendix Tables (11–30):

Clinical Domain	Benchmark Dataset	Key Pathologies / Findings	Top Reported AUC
Thoracic Radiography	LT-CXR (Table 11)	Subcutaneous Emphysema, Tortuous Aorta, Hernia, Pneumoperitoneum, Cardiomegaly	0.963, 0.962, 0.961, 0.954, 0.927
Hospital CXR Diagnoses	MGB CXR (Tables 15–16)	Post-resection volume loss, Tension pneumothorax, Spinal fixation, Sternal wires, Congestion	0.998, 0.996, 0.997, 0.993, 0.993
Echocardiography	Bing & HMC-QU (Tables 19–20)	CW-Doppler, M-mode Doppler, 2-Chamber View, Parasternal Long Axis View	0.995, 0.992, 1.000, 0.943
Dermatology Dermoscopy	ISIC 2019 (Table 21)	Dermatofibroma, Basal Cell Carcinoma, Vascular Lesion, Melanocytic Nevus, Melanoma	0.988, 0.985, 0.983, 0.957, 0.937
Clinical Dermatology	PAD-UFES-20 (Table 26)	Malignant Melanoma (MEL), Nevus (NEV), Basal Cell Carcinoma (BCC)	0.993, 0.986, 0.939
Retinal OCT	OCT2017 & 2018 (Tables 28–29)	Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), Drusen, Normal	1.000, 1.000, 1.000, 1.000 (100%)
Retinal Fundus	ODIR-5K (Table 30)	Diabetic Retinopathy, Glaucoma, Cataract, Age-related Macular Degeneration (AMD)	0.949 (Mean AUC)
Breast Ultrasound	BUSI (Table 17)	Normal breast tissue, Benign fibroadenoma, Malignant breast lesions	0.985 (Mean AUC)

Conclusion & Clinical Translation:

MedImageInsight demonstrates that high-performance, general-domain medical foundation models do not require prohibitive parameter scales (e.g. 7B–84B) or single-domain specialization. By unifying DaViT-based vision encoding with UniCL multi-domain alignment, it provides instantaneous zero-shot classification, inherently transparent KNN search for evidence retrieval, and full regulatory ROC compliance across diverse clinical disciplines.